DSC5001 Project Proposal

Team [Team Name] · Submitted [DD Month 2026]

Project title	[A specific, question-style title, e.g. What Drives X in Y? A Statistical Learning Study]
Team name	[Team Name]

Member	UID	Main responsibility
[Name 1]	[UID]	[Data collection & cleaning]
[Name 2]	[UID]	[Baseline & linear models]
[Name 3]	[UID]	[Non-linear / ensemble models]
[Name 4]	[UID]	[Interpretation & robustness]
[Name 5 (if any)]	[UID]	[Presentation & project management]

1. Background and Motivation

2 to 4 sentences. Say what the problem is, who cares about the answer, and what decision it would inform. The course grades scientific value: show that this is a problem you genuinely care about.

[Problem context]. [Who is affected / who would use the result]. [Why existing knowledge or practice is not enough]. This project aims to [one-sentence objective].

2. Research Questions

2 to 3 specific questions that data can answer or refute. Avoid 'we will predict X' with no reason.

- RQ1: [a question answerable with the data]
- RQ2: [a question answerable with the data]
- RQ3: [a question answerable with the data (optional)]

3. Data

Name the source and link, the time span or snapshot date, the unit of observation, approximate size, the response variable and the main groups of predictors. Mention known data issues.

- **Source:** [dataset name, provider, URL, access date]
- **Unit and size:** [each row is ...; about N rows and p variables]
- **Response:** [variable, type (continuous / binary / count), any transformation]
- **Predictors:** [main groups, plus engineered features you plan to build]
- **Known issues:** [missing values, imbalance, outliers, possible leakage]

4. Proposed Methods

Use a ladder from simple to flexible so each step can be justified against the one before. Link each method to a research question.

- **Baseline:** [naive benchmark, e.g. group mean / majority class]
- **Interpretable models:** [e.g. linear or logistic regression; Ridge / Lasso]
- **Flexible models:** [e.g. GAM / splines; random forest; gradient boosting; SVM]
- **Unsupervised (optional):** [e.g. PCA, k-means, hierarchical clustering]

5. Validation and Evaluation

Explain how you will avoid over-fitting and leakage. Keep the test set untouched until the end.

- **Split:** [e.g. 80/20 train/test with fixed seed; time-based split for time series]
- **Tuning:** [e.g. 10-fold cross-validation on the training set]
- **Metrics:** [e.g. RMSE, MAE, R-squared / AUC, F1, calibration] with [bootstrap confidence intervals]
- **Interpretation:** [e.g. coefficients, permutation importance, partial dependence plots]
- **Robustness:** [e.g. sub-samples, alternative cleaning rules, removing top feature]

6. Expected Contribution

What new or useful perspective will the conclusions offer, and to whom? State limitations you already foresee.

We expect to [main finding you will test]. The results could help [user] to [decision]. Limitations: [e.g. observational data, single snapshot, correlation not causation].

7. Timeline

Final submission deadline: 29 Nov 2026, 11:59 PM. Leave at least one day of buffer.

Week	Dates	Milestone
1	[dates]	[Data collection, cleaning, EDA]
2	[dates]	[Baseline and main models with cross-validation]
3	[dates]	[Model comparison, interpretation, robustness; freeze results]
4	[dates]	[Slides (max 10 pages), 8-min recording, code appendix, submit]

References

Cite the dataset and any paper or method source you rely on.

[1] [Dataset citation with URL and access date]

[2] [Related study or textbook, if any]